Notes: Danzer and Zyska (AEJ: Economic Policy, 2023)

Pensions and Fertility: Microeconomic Evidence

Contents

1 Big picture

- **Question.** Does expanding old-age pensions reduce fertility? More specifically, when households receive a large exogenous rise in future pension wealth, do women reduce childbearing because children become less necessary as providers of old-age support?
- **Setting.** The paper studies Brazil's 1991 pension reform, which effectively equalized pension rules between rural and urban workers. The reform mattered most for rural workers, especially women, because rural women had been largely excluded from meaningful pension coverage before 1991.
- **Core challenge.** Rural fertility was already falling. So a simple before-after comparison would mix the reform with broader changes in education, health care, religion, mass media, trade liberalization, stabilization policy, and long-run modernization.
- **Main conceptual contribution.** The paper brings the old-age-security hypothesis of fertility to micro data. Instead of asking whether countries with bigger pension systems have lower fertility, it asks whether a sharp, policy-driven rise in pension wealth changes actual birth behavior of affected women.
- **Main hypotheses.** The reform should reduce fertility most strongly for women who are older, closer to completing fertility, already have several children, and already have sons. Older women should respond more because they have a shorter discount horizon, more information about their children, and are nearer to their fertility stopping point.
- **Main empirical design.** The short-run analysis is a difference-in-differences design comparing rural women (treated) to urban women (control) before and after 1991. The paper then uses the rural-post interaction as an instrument for pension wealth in a 2SLS design, and finally estimates an event study for completed fertility at age 45 in the long run.
- **Main short-run result.** The reform lowers the annual probability of childbirth for women ages 15–44 by about 0.9 percentage points, roughly an 8% decline. The effect is concentrated among women ages 30–44, whose childbearing probability falls by about 2.4 percentage points, or about 24%.
- **Main heterogeneity result.** The effect is essentially absent for younger women ages 15–29 in the short run, but becomes stronger at higher birth parities and among women who already have sons.
- **Main long-run result.** Completed fertility at age 45 falls by about 1.3 children within roughly twenty years after the reform. The event-study graph shows no meaningful pre-trend break and then a sustained post-1991 decline in rural relative to urban completed fertility.
- **Main mechanism interpretation.** The evidence is most consistent with children serving partly as informal old-age insurance. Once generous public pensions arrive, the desired number of children falls, especially for women who are already late in the fertility cycle.
- **Main credibility result.** The paper works hard to rule out alternative explanations: macro stabilization, education, health, religion, telenovelas, trade shocks, migration, household income, wealth, and labor supply trends do not line up with the estimated rural-relative fertility decline.
- **Broader takeaway.** Pension policy changes not only redistribution and retirement behavior; it also changes family formation and the long-run demographic base of a pay-as-you-go system. In this paper, the fertility effect is large enough to matter for the system's fiscal sustainability.

2 Data and measurement (key objects)

2.1 The policy reform and why Brazil is a useful setting

Brazil is a powerful setting because, before the reform, rural workers were only weakly covered by the pension system, while urban workers already had relatively mature coverage. The 1991 reform therefore generated a large rise in pension generosity for rural workers and only small changes for urban workers.

The key rural changes were:

- old-age pension eligibility fell to age 55 for women and 60 for men,
- the minimum benefit rose from 50% to 100% of the minimum wage,
- the one-beneficiary-per-household restriction was abolished,
- and even unpaid or informal rural workers could qualify under extremely soft proof-of-work rules.

Urban workers, by contrast, saw almost no comparable reform in eligibility. The main urban change was that the minimum benefit rose from 90% to 100% of the minimum wage. This asymmetry is exactly what the paper uses.

A simple institutional summary from Table 1 is therefore:

- **Rural before 1991:** meager, exclusionary, and effectively male-centered.
- **Rural after 1991:** broad, generous, and much more individually based.
- **Urban before and after 1991:** already covered, with much smaller marginal change.

Interpretation. The reform looks like a large pension-wealth shock for rural women, not a general Brazilian fertility policy.

2.2 Evidence that the reform was economically large

The paper documents three first-stage facts.

First, rural eligibility and receipt exploded after 1991. Rural pension eligibility more than doubled between 1990 and 1992, and the number of recipients rose with a short administrative lag.

Second, pension income rose much more in rural areas. Between 1990 and 1992, average monthly pension income increased by about 15% for urban pensioner households but by about 134% for rural pensioner households.

Third, the relevant wealth object for the fertility analysis—the present value of future pension income for not-yet-retired workers—rose dramatically. Figure 3 shows that pension wealth of rural worker couples increased by roughly a factor of three, while the change for urban workers was much smaller.

Interpretation. The paper is not identifying on a tiny institutional tweak. It is identifying on a genuinely large shift in expected old-age resources.

2.3 Data source and sample construction

The main data source is the **Pesquisa Nacional por Amostra de Domicilios (PNAD)**, the annual Brazilian National Sample Household Survey. The paper uses repeated cross-sections from 1981–1990, 1992–1993, 1995–1999, 2001–2009, and 2011–2014. The PNAD was not conducted in 1980, 1991, 1994, 2000, or 2010.

The paper builds two main samples:

- **Short-run sample:** women ages 15–44 in 1981–1999, excluding the 2000s because later pension and health reforms would contaminate short-run identification. Sample size: about 1.44 million observations.

- **Long-run sample:** women from birth cohorts 1930–1969 observed at ages 45–69 in survey years that contain fertility histories. Their completed fertility is assigned to the year in which they were age 45. Sample size: about 746,671 observations.

2.4 Treatment and control groups

Treatment is defined by *occupation*, not by residence. This matters because Brazil's pension system enrolled people by labor-market category rather than by where they lived.

The rural/urban classification is constructed hierarchically:

1. current occupation in the reference week,
2. retrospective personal occupation up to four years earlier,
3. occupation of the household head,
4. household location as a last resort.

In the short-run sample, about 45% of observations are classified from own current occupation, 6% from own past occupation, 42% from the occupation of the family head, and 7% from residence. Mixed rural-urban couples are excluded.

Interpretation. This design weakens the concern that selective residential migration drives the result. The paper is trying to capture pension regime exposure, not simply rural living.

2.5 Outcome variables

The paper uses two fertility outcomes.

Short-run childbearing probability. A dummy equal to one if a woman gave birth within the last 12 months, and zero otherwise.

Long-run completed fertility. The total number of births a woman has had by age 45.

This split is important. The short-run outcome detects immediate behavioral responses. The completed-fertility outcome shows whether the reform actually changed the lifetime number of children rather than merely shifting birth timing.

2.6 Pension wealth

The paper computes age-specific gross present pension wealth as the discounted expected value of future pension payments:

$$\text{PensionWealth}_a = \sum_{t=0}^{T-a} s_{a,t} \frac{1}{(1+i)^t} \text{pension}_t, \tag{1}$$

where $s_{a,t}$ is the survival probability from current age a to year t, $T-a$ is remaining maximum lifespan, and i is the real discount rate. The baseline discount rate is 12%.

The intuition is simple: older women gain more pension wealth from the same reform because they are closer to retirement and discount the future over fewer years.

2.7 Main controls

The DID and IV specifications include rich controls:

- individual controls: age dummies, years of schooling, marriage, prior children,

- job controls: hours worked, employment status, woman's income share,
- household controls: income excluding pensions, wealth, adults in household, potential caretaker, partner age,
- regional and group controls: trade shocks, TV coverage, religion shares, race shares, and macroregional child mortality.

Interpretation. The fixed effects do most of the identification work, but the control set is designed to soak up many channels through which rural and urban fertility trends might otherwise have diverged.

2.8 Descriptive facts

Table 2 already shows the basic pattern.

In the short-run sample, rural women had higher fertility than urban women before the reform and remained less educated and poorer, but the rural childbearing rate fell from 0.14 to 0.11 while the urban rate fell from 0.09 to 0.07.

In the long-run sample, completed fertility fell from 6.83 to 5.25 births for rural women and from 4.63 to 3.08 births for urban women between the pre- and postreform samples.

Interpretation. Rural and urban women were always different in levels. That is why the panel structure and fixed-effects logic are essential: the identification is based on changes in the rural-urban gap, not on raw differences.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper starts from the old-age-security hypothesis of fertility. Children provide utility, but they can also serve as an informal insurance device for old age. A pension expansion raises the value of formal old-age support and therefore lowers the need to rely on future transfers from children.

The paper organizes heterogeneity around three dimensions:

- **Age.** Older women should react more in the short run because they are nearer retirement, closer to fertility completion, and have more information about the survival and quality of their children.
- **Birth parity.** If desired fertility shifts downward, the strongest negative effects should appear at higher parities rather than on the first child.
- **Gender of older children.** Because sons are relatively important in Brazilian inheritance practices, women who already have sons should react more strongly to the pension reform.

3.2 Baseline DID specification

The main short-run estimating equation is

$$y_{igtr}^{SR} = \alpha^{SR} + \beta\big(RURAL_g \times POST_t\big) + \gamma RURAL_g + \lambda_t + \phi_r + \psi' X_{igtr}^{SR} + \varepsilon_{igtr}^{SR}. \tag{2}$$

Here:

- y_{igtr}^{SR} is a dummy for whether woman i gave birth in the last 12 months,

- $RURAL_g$ indicates belonging to the rural treatment group,
- $POST_t$ indicates the post-1991 reform period,
- λ_t are year fixed effects,
- ϕ_r are region fixed effects,
- and X_{igtr}^{SR} is the large control vector.

Interpretation of β. It is the differential change in childbearing for rural women after the reform relative to urban women after netting out common time effects and time-invariant regional differences. A negative β means the pension reform lowered fertility.

3.3 Why the DID is appealing here

The DID design is attractive because the reform was sharp, national, and institutionally tied to the rural pension regime. The control group—urban women—lived through the same macroeconomic environment but saw little change in pension wealth.

This means the main identifying assumption is a **parallel-trends assumption**: absent the reform, fertility among rural and urban women would have evolved similarly once common shocks and fixed differences are removed.

3.4 IV specification for the semi-elasticity of fertility with respect to pension wealth

The paper then asks not only whether the reform mattered, but how fertility responds to pension wealth more generally. The first stage is

$$\ln(\text{PensionWealth}_{igtr}) = \sigma^{SR} + \theta\big(RURAL_g \times POST_t\big) + \eta RURAL_g + \delta_t + \tau_r + \kappa' X_{igtr}^{SR} + \mu_{igtr}^{SR}, \tag{3}$$

and the second stage is

$$y_{igtr}^{SR} = \alpha^{SR} + \beta \ln\big(\widehat{\text{PensionWealth}}_{igtr}\big) + \gamma RURAL_g + \lambda_t + \phi_r + \psi' X_{igtr}^{SR} + \varepsilon_{igtr}^{SR}. \tag{4}$$

Interpretation. The IV coefficient β is a semi-elasticity: it tells us how the annual probability of childbirth changes when pension wealth rises by a given percentage.

The first stage is extremely strong. In the main sample the excluded instrument has an F-statistic above 4,100 and a t-statistic above 64.

3.5 Long-run event-study specification

To study completed fertility, the paper estimates

$$y_{igtr}^{LR} = \alpha^{LR} + \sum_{\tau=-16}^{23} \beta_\tau \big(RURAL_g \times YEAR_\tau\big) + \gamma RURAL_g + \lambda_t + \phi_r + \psi' X_{igtr}^{LR} + \varepsilon_{igtr}^{LR}. \tag{5}$$

Here y_{igtr}^{LR} is the total number of births at age 45. The coefficients on the leads test for pre-trends, and the coefficients on the lags show whether rural completed fertility falls relative to urban completed fertility after the reform.

Interpretation. This is the right tool for the long-run question because women differ in how much time they had left to adjust their lifetime fertility after 1991.

3.6 A simple accounting equation for the PAYG system

To connect the fertility estimates to pension finance, the paper writes the long-run funding effect of the reform as

$$\text{Gap} = -\big(\Delta B \times 13 \times \Delta N\big)(1 + AC) + \big(C \times 12\big)\Delta L, \tag{6}$$

where ΔB is the increase in average benefits, ΔN the rise in beneficiaries, C monthly contributions per worker, ΔL the loss of future contributors from lower fertility, and AC administrative cost.

Interpretation. The reform worsens the fiscal balance from two sides: more generous benefits and fewer future contributors.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the short-run effect

Identification comes from the fact that rural women experienced a much larger increase in pension wealth than urban women after 1991. The DID coefficient compares changes in the rural-urban fertility gap before and after that date.

A useful way to say it is: the paper asks whether rural childbearing fell more than urban childbearing exactly when the pension system suddenly became much more valuable for rural women.

4.2 Why anticipation is limited

The paper argues that anticipation is weak. Although the 1988 constitution opened the door to reform, the exact design and timing of implementation remained uncertain until 1991. Qualitative evidence suggests that rural households were poorly informed about the concrete consequences before benefits actually started to arrive. This matters because if households had fully anticipated the reform years earlier, fertility would have started adjusting before 1991.

4.3 Parallel trends and placebo reforms

The paper directly checks pre-trends. In the pre-1991 period, almost all rural-year interactions are near zero, and only 2 of 27 are significant at the 10% level, which is fully consistent with ordinary Type I error.

The paper also runs placebo-reform exercises at other dates and finds no evidence of comparable fertility responses before the actual reform.

Bottom line. The design does not rely on a blind leap of faith; the pretrend evidence is explicitly shown.

4.4 Why migration and composition are not driving the results

Because treatment is assigned mainly by occupation rather than residence, selective residential migration is less threatening than in a purely geographic design. The paper also notes that interstate migration was modest in the relevant period—below 8% for urban residents and 5% for rural residents over eight years.

The authors also check for compositional changes, such as rural women becoming more likely to hold urban occupations after the reform, and do not find meaningful evidence of this.

4.5 Alternative confounders the paper addresses

The paper carefully reviews rival explanations and shows why they are unlikely to generate the main pattern.

Macroeconomic stabilization. Brazil had major stabilization plans in the late 1980s and early 1990s, but the paper argues these did not differentially improve the relative position of rural women in a way that matches the fertility response.

Education. Schooling rose over time for both rural and urban women, but the trajectories were similar. The big expansion in compulsory schooling also came with long implementation delays.

Health and child mortality. Child mortality fell more in urban than in rural areas between 1986 and 1996. If anything, that would bias the estimated rural fertility decline downward, not upward.

Women's empowerment. The female income share in household income does not show a rural-relative jump that could explain the results.

Labor supply. Labor-force participation trends are similar between rural and urban women. Rural working hours actually fall disproportionately after the reform because of part-time jobs, which would tend to reduce the opportunity cost of childbearing and therefore work against the paper's main finding.

Income and wealth. Household income excluding pensions and household wealth do not show a post-1991 rural-relative break that lines up with the fertility effect.

Religion and culture. The rise of Protestantism and secularization is actually more pronounced in urban areas, not rural ones. Telenovelas are controlled for through Rede Globo coverage, and their spread occurred mainly earlier.

Trade shocks. The regressions incorporate region-industry trade shocks, and women in rural and urban jobs experienced similar tariff-exposure trends.

4.6 What the IV identifies and what it does not

The IV design uses the rural-post interaction as an instrument for pension wealth. This gives a causal elasticity under the assumption that the reform affects fertility only through pension wealth, not through some other independent rural-specific channel.

The paper argues this exclusion restriction is plausible because the contribution side of the system did not change for workers and the main treatment was the large rise in pension generosity.

Still, the IV should be interpreted as a policy-induced semi-elasticity around this specific reform environment, not as a universal structural parameter for every pension system in every country.

4.7 Main limitation

The paper is very strong on causal identification of the fertility response to this Brazilian pension reform. It is less able to fully separate all possible motives behind the response: old-age security, changed expectations about intrafamily support, intra-household bargaining, and broader dynamic family planning all sit underneath the same revealed behavior.

That is a limitation, but not a fatal one. The paper's main claim is about the existence and size of the fertility response, and on that dimension the evidence is convincing.

4.8 Relation to the literature

The paper contributes to three literatures at once:

- the macro and cross-country literature linking pensions and fertility,

- the development literature on social insurance expansion,
- and the demographic literature on old-age security motives.

Its main innovation is to move from broad correlations to a clean quasi-experimental micro design in a developing-country setting where old-age family support is especially relevant.

5 Tables and figures

5.1 Table 1 and Figures 1–3: what changed in the reform

TABLE 1—THE BRAZILIAN RETIREMENT PENSION SYSTEM BEFORE AND AFTER THE 1991 REFORM

Occupation	Pension type		Pension system before the reform	Pension system after the reform
Rural occupation	Old age	Eligibility criteria:	Minimum age of 65 years, documented formal employment for one out of the past three years or three years overall. Weak documentation requirements.	Minimum age of 60/55 for men/women.
		Benefit level:	Flat and equal to 50 percent of the minimum wage.	Same rules as urban workers.
		Restrictions:	Only one person per household is eligible.	No restriction in the number of beneficiaries per household.
	Length of service	Eligibility criteria:	Not eligible.	Same rules as urban workers.
	Disability	Eligibility criteria:	Available at any age.	Available at any age.
		Benefit level:	Flat and equal to 50 percent of the minimum wage.	Same rules as urban workers.
		Restrictions:	Need to stop working.	Need to stop working.
Urban occupation	Old age	Eligibility criteria:	Minimum age of 65/60 for men/women.	Same age limits as before.
		Benefit level:	Benefit level is 90 percent of the minimum wage. Earnings-related part determined by length-of-service pension.	Benefit is 70 percent of earnings-based benefit (average income over last 36 months) at the minimum eligibility age plus 1 percent for each year of payroll tax contributions, up to 100 perecent. Minimum benefit increased to 100 percent of the minimum wage.
		Restrictions:	Need to quit the current job to apply for benefits. No earnings or retirement tests after that.	Allowed to stay in the current job.
	Length of service	Eligibility criteria:	Eligibility after 30 years of work. Full benefits after 35 years of work. Fewer years for some types of work. No minimum age requirements.	Eligibility after 30/25 years of work for men/women. Full benefits after 35/30 years of work for men/women. No minimum age requirements.
		Benefit level:	Benefits determined by years of documented work and recent labor earnings. Minimum benefit is 90 perceent of the minimum wage. Bonus for continued work beyond maximum eligibility period.	Benefit is 70 percent of earnings-based benefit (average income over last 36 months) at the minimum eligibility age plus 6 percent for each additional year of service, up to 100 percent. Minimum benefit increased to 100 percent of the minimum wage.
		Restrictions:	Need to quit current job to apply for benefits. No earnings or retirement test after that.	
	Disability	Eligibility criteria:	Available at any age.	Available at any age.
		Benefit level:	Benefit is 90 percent of minimum wage.	Benefit is 80 percent of earnings-based benefit plus 1 percent for each year of payroll tax contributions, up to 100 percent. Minimum benefit level of 100 percent of the minimum wage.
		Restrictions:	Need to stop working.	Need to stop working.

Notes: After the reform, monthly benefit levels for all types of pensions (old age, length of service, and disability) were equalized for all eligible individuals and formally determined as follows: $OldAge_i = max\{[0.7 + 0.01 \times (P_i/12)] \times B_i, 1 \times minwage\}$, $LengthofService_i = max\{(0.7 + 0.06 \times ADS_i) \times B_i, 1 \times minwage\}$, $Disability_i = max\{[0.8 + 0.01 \times (P_i/12)] \times B_i, 1 \times minwage\}$, with B_i being the benefit income determined as the average of the final 36 monthly earnings on which individual i has paid social security contributions. P_i indicates the number of months during which individual i paid social security contributions (payments of the payroll tax) during her entire career. ADS_i indicates additional years of service. The minimum benefit level—which is also applicable for all individuals who fail to prove formal earnings or social security contributions—equals 100 percent of the minimum wage for all types of pensions. The pension system makes 13 benefit payments per year, which are uprated along with the national CPI; the minimum pension can be uprated by more than the CPI.

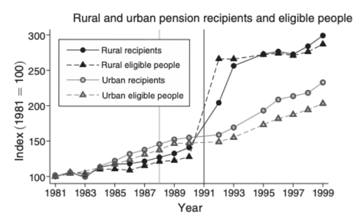

FIGURE 1. DEVELOPMENT OF THE INDEXED NUMBER OF PENSION-ELIGIBLE PEOPLE AND RECIPIENTS (1981 = 100), BRAZIL, 1981–99

Notes: The graph shows the indexed number of rural and urban people eligible for an old-age pension and old-age pension recipients (1981 = 100). Vertical lines indicate the new constitution, approved in 1988 (light gray), and the pension reform implemented in 1991 (black). Eligibility and receipt are defined by age: 65 for rural elderly before the reform, 55/60 for women/men after the reform, and 60/65 for urban women/men before and after the reform. Rural and urban groups are defined according to occupation and location (for details, see Section III).

Source: PNAD 1981–1990, 1992–1993, 1995–1999

Read:

- Table 1 shows why the treatment is so strong for rural women. Before the reform, rural pensions were minimal, poorly covered, and effectively constrained to one household member. After the reform, rural women could qualify at age 55 and receive a benefit floor equal to the full minimum wage.
- Figure 1 shows that the number of rural pension-eligible people and recipients jumps sharply after 1991, while the urban series is comparatively smooth.
- Figure 2 shows that average monthly pension income rises by about 134% in rural pensioner households between 1990 and 1992, versus only about 15% in urban households.
- Figure 3 shows that gross present pension wealth rises far more for rural than urban women and couples;

for rural worker couples the increase is roughly a factor of three.

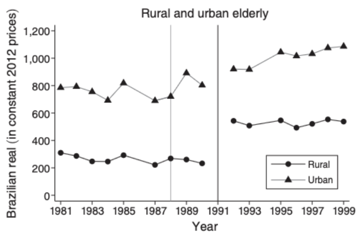

FIGURE 2. AVERAGE MONTHLY PENSION INCOME, BRAZIL, 1981–1999

Notes: The graph shows the average household pension income of rural and urban households with at least one old-age pension recipient, based on eligibility by age. Vertical lines indicate the new constitution, approved in 1988 (light gray), and the pension reform implemented in 1991 (black). Rural and urban groups are defined according to household location.

Source: PNAD 1981–1990, 1992–1993, 1995–1999

FIGURE 3. GROSS PRESENT PENSION WEALTH OF RURAL AND URBAN FEMALE WORKERS AND RURAL AND URBAN WORKER COUPLES WITH A WOMAN AGE 15–44

Notes: Pre- and postreform pension wealth is computed as the present value of expected old-age benefits before and after the reform, adjusted for real interest rates and average survival probabilities (computed using IBGE mortality tables, available for the first time in 1998): $PensionWealth = \sum_{t=0}^{T-a} s_{a,t} \times \left[1/(1+i)^t\right] \times pension_t$, with $s_{a,t}$ denoting the probability of a person of age a in a given year surviving until year t; $T - a$ indicates the remaining maximum lifespan, differentiated by sex and birth cohort; i is a constant discount rate (12 percent, following Azzoni and Isai 1994); and $pension_t$ denotes the old-age pension benefits in t. A nonretired person receives the pension starting in a future period $t > 0$, defined by the person's age and the regular retirement age. Rural and urban groups are defined according to occupation and location (for details, see Section III). Online Appendix Figure A2 shows the pension wealth of rural and urban male workers. For further details on the pension wealth computation, see online Appendix Section B.

Source: PNAD 1981–1990, 1992–1993, 1995–1999

Economic message. The reform generated a very large first stage for rural women. Without Figures 1–3, the fertility results would just look like reduced-form correlations. With them, it is clear the paper is studying a major pension-wealth shock.

5.2 Table 2 and Figure 4: descriptive patterns and confounder trends

Read:

- Table 2 shows that rural women start with higher fertility, lower schooling, lower income, and lower wealth than urban women both before and after the reform.
- In the short-run sample, the fraction with a newborn falls from 0.14 to 0.11 in rural occupations and from 0.09 to 0.07 in urban occupations.
- In the long-run sample, completed fertility falls from 6.83 to 5.25 births for rural women and from 4.63 to 3.08 births for urban women.
- Figure 4 shows that marriage, education, child mortality, women's income share, labor-force participation, and working-hour trends do not display a post-1991 rural-relative break that obviously mirrors the fertility result.

Economic message. Rural and urban women differ in levels, but the confounder trends do not show the right timing or direction to explain the main estimates.

5.3 Figure 5 and Table 3: the main DID result

Read:

- Figure 5 shows that rural and urban fertility trends move similarly before 1991 and then diverge after the reform, especially for women ages 30–44.

TABLE 2—DESCRIPTIVE STATISTICS FOR PRE- AND POSTREFORM SAMPLES

Panel A. Sample A: Short-run analysis

	Pre-reform (N = 740,994)				Postreform (N = 471,623)			
Urban occupation	Mean	SD	Min.	Max.	Mean	SD	Min.	Max.
Newborn child (under one year old)	0.09	0.29	0	1	0.07	0.25	0	1
Age	27.36	8.29	15	44	28.20	8.52	15	44
Age of partner	31.92	9.16	15	99	32.41	9.30	16	98
Years of education	6.40	4.07	0	17	7.36	3.98	0	17
Married	0.52	0.50	0	1	0.52	0.50	0	1
Household income	3,000	4,892	0	820,542	2,509	4,104	0	564,481
Woman's income share	0.15	0.27	0	1	0.20	0.31	0	1
Wealth index	0.17	0.96	−8.69	9.31	0.14	0.91	−8.55	12.13
Not worked reference week	0.53	0.50	0	1	0.49	0.50	0	1
Worked 1–10 hours ref. week	0.01	0.10	0	1	0.02	0.13	0	1
Worked 11–20 hours ref. week	0.05	0.21	0	1	0.06	0.23	0	1
Worked 21–30 hours ref. week	0.07	0.25	0	1	0.07	0.26	0	1
Worked 31–40 hours ref. week	0.13	0.34	0	1	0.15	0.36	0	1
Worked 41–50 hours ref. week	0.15	0.35	0	1	0.15	0.36	0	1
Worked 51–60 hours ref. week	0.04	0.20	0	1	0.04	0.20	0	1
Worked > 60 hours ref. week	0.02	0.15	0	1	0.02	0.14	0	1
Number of adults in household	2.99	1.50	0	22	2.82	1.38	0	16
Caretaker in household	0.17	0.37	0	1	0.16	0.37	0	1

	Pre-reform (N = 158,317)				Postreform (N = 71,442)			
Rural occupation	Mean	SD	Min.	Max.	Mean	SD	Min.	Max.
Newborn child (under one year old)	0.14	0.35	0	1	0.11	0.31	0	1
Age	27.08	8.84	15	44	27.53	8.92	15	44
Age of partner	32.58	10.35	15	98	32.91	10.49	15	89
Years of education	2.69	2.68	0	17	3.53	2.96	0	17
Married	0.62	0.49	0	1	0.62	0.49	0	1
Household income	1,236	2,704	0	183,294	893	2,157	0	176,907
Woman's income share	0.05	0.17	0	1	0.06	0.19	0	1
Wealth index	−0.24	0.96	−7.93	8.57	−0.23	1.12	−7.31	9.98
Not worked reference week	0.72	0.45	0	1	0.68	0.47	0	1
Worked 1–10 hours ref. week	0.00	0.02	0	1	0.00	0.05	0	1
Worked 11–20 hours ref. week	0.02	0.15	0	1	0.08	0.27	0	1
Worked 21–30 hours ref. week	0.06	0.25	0	1	0.08	0.27	0	1
Worked 31–40 hours ref. week	0.08	0.28	0	1	0.07	0.26	0	1
Worked 41–50 hours ref. week	0.07	0.25	0	1	0.06	0.23	0	1
Worked 51–60 hours ref. week	0.03	0.17	0	1	0.02	0.15	0	1
Worked > 60 hours ref. week	0.01	0.09	0	1	0.01	0.10	0	1
Number of adults in household	2.88	1.37	0	15	2.78	1.31	0	16
Caretaker in household	0.14	0.35	0	1	0.15	0.36	0	1

Panel B. Sample B: Long-run analysis

	Pre-reform (N = 207,388)				Postreform (N = 418,982)			
Urban occupation	Mean	SD	Min.	Max.	Mean	SD	Min.	Max.
Total number of births	4.63	3.64	0	20	3.08	2.43	0	20
Childlessness (number of births = 0)	0.11	0.31	0	1	0.11	0.31	0	1
Years of education	4.46	4.37	0	17	7.09	4.88	0	17
Married	0.51	0.50	0	1	0.60	0.49	0	1
Lifetime labor supply (years)	13.91	1.24	10.36	16.73	16.22	1.50	10.03	20.29

	Pre-reform (N = 52,175)				Postreform (N = 68,126)			
Rural occupation	Mean	SD	Min.	Max.	Mean	SD	Min.	Max.
Total number of births	6.83	4.33	0	20	5.25	3.68	0	20
Childlessness (number of births = 0)	0.08	0.27	0	1	0.07	0.26	0	1
Years of education	1.68	2.50	0	17	2.64	3.12	0	17
Married	0.71	0.45	0	1	0.79	0.41	0	1
Lifetime labor supply (years)	8.80	2.84	0.05	15.87	10.27	3.06	0.21	17.81

Notes: Sample A consists of Brazilian women ages 15–44. Sample B consists of Brazilian women in the birth cohorts from 1930 to 1969, ages 45–69. Rural and urban groups are defined by occupation during the reference week, occupation up to four years prior to the reference year, occupation of the family head, and household location. Mixed urban-rural couples are excluded. For more information, see the variable descriptions in online Appendix Tables A6–A11.

Sources: Sample A: PNAD 1981–1990, 1992–1993, 1995–1999; Sample B: PNAD 1984–1985, 1992–1993, 1995–1999, 2001–2009, 2011–2014

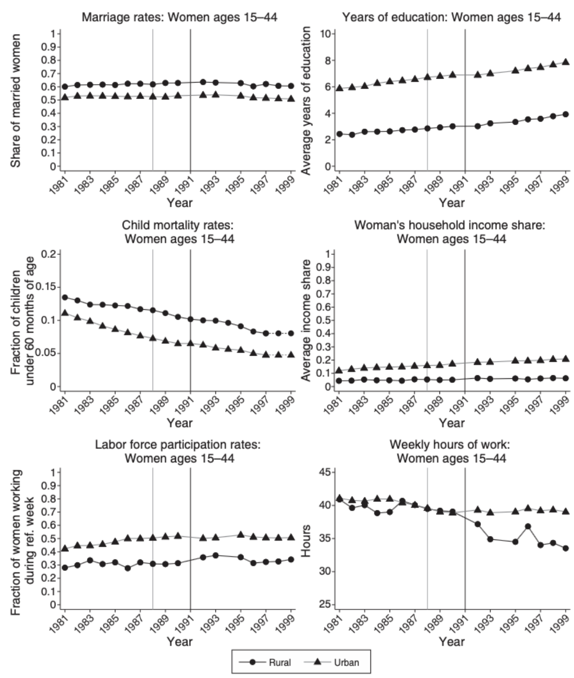

FIGURE 4. CONFOUNDING FACTOR TRENDS: WOMEN AGES 15–44

Notes: Graphs show average marriage rates, years of education, child mortality rates, income of women as a share of household income, labor force participation rates, and weekly hours of work for rural and urban female workers. Vertical lines indicate the new constitution approved in 1988 (light gray) and the pension reform implemented in 1991 (black). Rural and urban groups are defined as in Table 2. Mixed urban-rural couples are excluded. Online Appendix Figure A9 shows the trends for women ages 15–29 and online Appendix Figure A10 for women ages 30–44.

Source: PNAD 1981–1990, 1992–1993, 1995–1999

- Table 3, column 6 is the preferred DID specification. The coefficient for all women ages 15–44 is −0.009, which is a 7.6% decline relative to baseline fertility.
- For women ages 15–29, the estimate is 0.003 and statistically unimportant.
- For women ages 30–44, the estimate is −0.024, which corresponds to about a 24.1% decline.

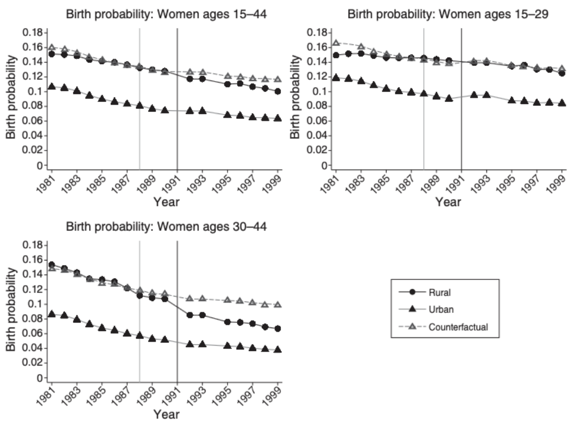

FIGURE 5. BIRTH PROBABILITIES: POOLED (WOMEN AGES 15–44) AND SUBGROUPS (WOMEN AGES 15–29 VERSUS 30–44), BRAZIL, 1981–1999

Notes: The graphs show three-year moving averages (two years at the edges: 1981–1982, 1989–1990, 1992–1993, and 1998–1999) of birth probabilities, i.e., average childbearing rates (zero or one) within the last 12 months. The counterfactual trend as of 1987 is represented by the gray, dashed line. Vertical lines indicate the new constitution approved in 1988 (light gray) and the pension reform implemented in 1991 (black). Rural and urban groups are defined as in Table 2. Mixed urban-rural couples are excluded.

Source: PNAD 1981–1990, 1992–1993, 1995–1999

TABLE 3—DID REGRESSION RESULTS: POOLED (WOMEN AGES 15–44) AND SUBGROUPS (WOMEN AGES 15–29 VERSUS 30–44)

Dependent Variable	Newborn child (under one year old): zero or one					
	(1)	(2)	(3)	(4)	(5)	(6)
DID women ages 15–44	−0.008	−0.008	−0.010	−0.008	−0.007	−0.009
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.003)
Percent to baseline	−6.8	−6.8	−8.3	−6.8	−6.0	−7.6
DID women ages 15–29	0.004	0.005	0.003	0.004	0.005	0.003
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.003)
Percent to baseline	2.9	3.6	2.2	2.9	3.6	2.2
DID women ages 30–44	−0.029	−0.029	−0.028	−0.026	−0.025	−0.024
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Percent to baseline	−27.7	−27.7	−27.0	−25.5	−24.8	−24.1
Year and region FE	No	Yes	Yes	Yes	Yes	Yes
Covariates (see note):						
Individual	No	No	Yes	Yes	Yes	Yes
Job	No	No	No	Yes	Yes	Yes
Household	No	No	No	No	Yes	Yes
Regional/Group	No	No	No	No	No	Yes
Observations: 15–44 (1,442,376), 15–29 (854,814), 30–44 (587,562)						
R^2 15–44:	0.006	0.008	0.092	0.098	0.106	0.107
R^2 15–29:	0.003	0.005	0.131	0.136	0.142	0.142
R^2 30–44:	0.011	0.015	0.045	0.048	0.057	0.058

Notes: The table shows DID estimates of the pension reform. The dependent variable is a dummy for whether a child was born in the last 12 months. Individual covariates include years of schooling, a dummy for marriage, age dummies, and dummies for one to five or more prior children (omitting zero). Job-related covariates include dummies for 1–10, 11–20, 21–30, 31–40, 41–50, 51–60 and 60+ hours of professional work in reference week (zero hours—i.e., not working—is omitted) and share of household income (excluding pensions) earned by the woman. Household covariates include the log of monthly household income (excluding pensions), household wealth, dummies for the number of adults in the household, a dummy for the presence of a potential caretaker in the household (i.e., a nonworking pensioner age 60 or older), and dummies for the age of the woman's partner. Regional and group covariates include a dummy for Rede Globo coverage in the past year (based on La Ferrara, Chong, and Duryea 2012b), regional shares of religious affiliations (i.e., Protestant, other faiths, and no religion, omitting Catholic, based on census data), regional race shares (i.e., Black, mixed-race and others including Asian, Indigenous, and other, omitting White, based on census data), macroregional child mortality (mortality rates of children under 60 months of age by macroregion—i.e., northeast, midwest, south, and southeast, based on DHS data), and regional, industry-specific trade shocks (based on the methodology of Dix-Carneiro and Kovak 2017 and annual tariff rates taken from Paiva Abreu 2004). Rural and urban groups are defined as in Table 2. Standard errors clustered at the regional level are in parentheses.

Source: PNAD 1981–1990, 1992–1993, 1995–1999

Economic message. The average short-run fertility effect is real, but it is almost entirely an older-woman effect. That age pattern is exactly what the old-age-security mechanism predicts.

5.4 Table 4: IV elasticity with respect to pension wealth

Read:

- The 2SLS coefficient on log pension wealth is about −0.005 for the full sample.
- For younger women it is essentially zero (0.001).
- For older women it is much more negative (−0.012).
- The first stage is extremely strong: the excluded instrument has F-statistics of 4,135 in the full sample, 3,683 for ages 15–29, and 5,600 for ages 30–44.

Economic message. The IV confirms the DID rather than overturning it. Pension wealth itself is the key object, and older women's fertility is much more sensitive to it.

5.5 Figure 6 and Table 5: the effect by birth parity

Read:

- Figure 6 shows little or no postreform decline for parity 0 and parity 1, but increasingly clear declines at higher parities.
- Table 5 confirms this pattern. The coefficients are 0.009 at parity 0 and 0.005 at parity 1, both insignificant

Table 4—IV Regression Results: Semi-elasticity of Birth Probabilities to Pension Wealth

Dependent Variable	Newborn child (under one year old): zero or one		
	Women ages 15–44 (1)	Women ages 15–29 (2)	Women ages 30–44 (3)
Log of pension wealth	−0.005	0.001	−0.012
	(0.002)	(0.002)	(0.002)
First-stage statistics			
Dependent variable	Pension wealth of woman and her partner		
Instrument	Rural work × After reform interaction		
Coefficient	1.944	1.874	2.080
	(0.030)	(0.031)	(0.028)
F-test	4,135	3,683	5,600
T-statistic of excl. instrument	64.31	60.69	74.83
Partial R^2 of excl. instrument	0.084	0.101	0.073
Estimates for alternative discount rates			
Log of pension wealth	−0.004	0.001	−0.012
30% pre-reform versus 12% postreform	(0.001)	(0.001)	(0.002)
Log of pension wealth	−0.003	0.001	−0.008
30% rural versus 12% urban	(0.001)	(0.001)	(0.001)
Year and region FE	Yes	Yes	Yes
Covariates (see note)	Yes	Yes	Yes
Observations	1,442,376	854,814	587,562
R^2	0.107	0.142	0.055

Notes: The table shows IV estimates of the pension reform. The first-stage regression is of log pension wealth on the instrument ($RURAL_k^{SR} \times POST_t^{SR}$) and all other covariates. The power of the first-stage regression is reported with *F*-statistics. The dependent variable in the second stage is a dummy for whether a child was born in the last 12 months. The full set of covariates is as in Table 3, column 6. Pension wealth is computed as in Figure 3 with a constant discount rate of 12 percent for the main specification (c.f. Azzoni and Isai 1994). Rural and urban group definitions are as in Table 2. Mixed urban-rural couples are excluded. Standard errors clustered at the regional level are in parentheses.

Source: PNAD 1981–1990, 1992–1993, 1995–1999

in the direction predicted by the theory.

- At parity 2 the coefficient is −0.012 (about a 9.8% decline), at parity 3 it is −0.022 (18.5%), at parity 4 it is −0.042 (28.6%), and at parity 5+ it is −0.041 (23.2%).

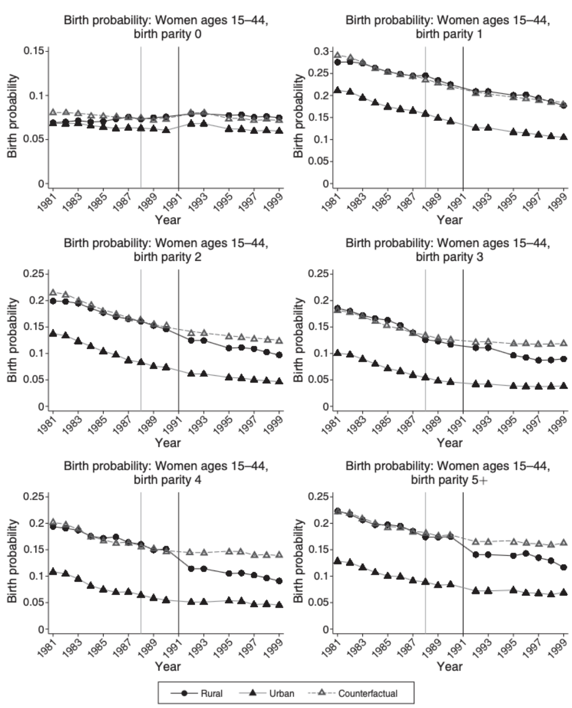

Figure 6. Birth Probabilities by Birth Parity

Notes: The graphs show three-year moving averages (two-year at the edges: 1981–1982, 1989–1990, 1992–1993, and 1998–1999) of birth probabilities, i.e., average childbearing (0/1) rates within the last 12 months by birth parity (the number of previous children) up to 5+. The counterfactual trend as of 1987 is represented by the gray, dashed line. Vertical lines indicate the new constitution approved in 1988 (light gray) and the pension reform implemented in 1991 (black). Rural and urban groups are defined as in Table 2. Mixed urban-rural couples are excluded.

Source: PNAD 1981–1990, 1992–1993, 1995–1999

Table 5—DID Regression Results by Birth Parity (Women Ages 15–44)

Dependent variable	Newborn child (under one year old): zero or one					
	Parity 0 (1)	Parity 1 (2)	Parity 2 (3)	Parity 3 (4)	Parity 4 (5)	Parity 5+ (6)
DID women ages 15-44	0.009	0.005	−0.012	−0.022	−0.042	−0.041
	(0.002)	(0.007)	(0.005)	(0.006)	(0.009)	(0.007)
Percent to baseline	10.4	2.5	−9.8	−18.5	−28.6	−23.2
Year and region FE	Yes	Yes	Yes	Yes	Yes	Yes
Covariates (see note)	Yes	Yes	Yes	Yes	Yes	Yes
Observations	682,684	222,497	232,969	147,617	70,809	85,800
R^2	0.199	0.078	0.071	0.074	0.078	0.064

Notes: The table shows DID estimates of the pension reform. The dependent variable is a dummy indicating whether a child was born in the last 12 months. The full set of covariates is as in Table 3, column 6 (dummies for zero to five or more prior children omitted). Rural and urban groups are defined as in Table 2. Mixed urban-rural couples are excluded. Standard errors clustered at the regional level are in parentheses.

Source: PNAD 1981–1990, 1992–1993, 1995–1999

Economic message. The reform does not mainly stop women from having a first or second child. It mainly reduces higher-order births. That is exactly what a downward shift in desired family size should look like.

5.6 Figure 7 and Table 6: the role of sons

Read:

- Figure 7 shows that fertility declines after the reform are concentrated among women who already have at least one son or at least two sons.
- Table 6 shows a DID estimate of −0.024 for women with at least one son and −0.032 for women with at least two sons, compared with an estimate near zero for women with only daughters.
- Among women ages 30–44, the effect is even larger: −0.030 for those with one or more sons and −0.037 for those with two or more sons.
- Women with no children do not reduce fertility.

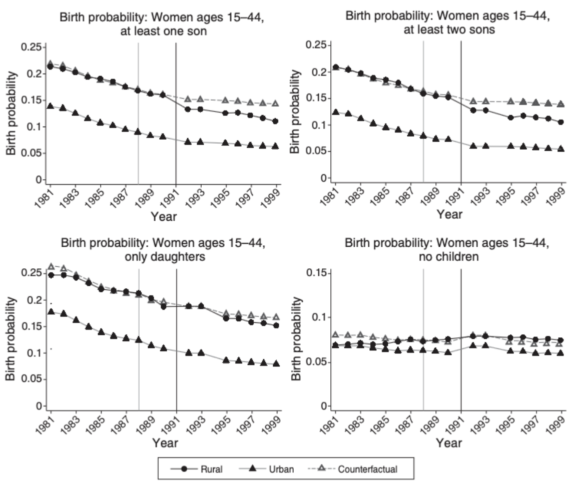

FIGURE 7. BIRTH PROBABILITIES BY OLDER CHILDREN'S GENDERS: WOMEN AGES 15–44

Notes: The graphs show three-year moving averages (two-years at the edges: 1981–1982, 1989–1990, 1992–1993, and 1998–1999) of birth probabilities, i.e., average childbearing (0/1) rates within the last 12 months, conditional on whether women had already given birth to at least one son, at least two sons, only daughters, or no children at all. The counterfactual trend as of 1987 is represented by the gray, dashed line. Vertical lines indicate the new constitution approved in 1988 (light gray) and the pension reform implemented in 1991 (black). Rural and urban groups are defined as in Table 2. Mixed urban-rural couples are excluded.

Source: PNAD 1981–1990, 1992–1993, 1995–1999

TABLE 6—DID REGRESSION RESULTS BY OLDER CHILDREN'S GENDER (WOMEN AGES 15–44)

Dependent variable	Newborn child (under one year old): zero or one				
	Full sample (1)	1+ sons (2)	2+ sons (3)	Only daughters (4)	No children (5)
DID women ages 15–44	−0.009	−0.024	−0.032	−0.002	0.009
	(0.003)	(0.005)	(0.005)	(0.006)	(0.002)
Percent to baseline	−7.6	−16.3	−21.5	−1.2	10.4
DID women ages 15–29	0.003	−0.008	−0.012	0.012	0.011
	(0.003)	(0.007)	(0.009)	(0.011)	(0.002)
Percent to baseline	2.2	−3.8	−5.4	4.9	11.7
DID women ages 30–44	−0.024	−0.030	−0.037	−0.012	0.006
	(0.004)	(0.005)	(0.006)	(0.006)	(0.003)
Percent to baseline	−24.1	−27.1	−29.5	−12.8	12.2
Year and region FE	Yes	Yes	Yes	Yes	Yes
Covariates (see note)	Yes	Yes	Yes	Yes	Yes
Observations 15–44	1,442,376	575,792	264,343	183,900	682,684
Observations 15–29	854,814	188,577	59,187	89,762	576,475
Observations 30–44	587,562	387,215	205,156	94,138	106,209
R^2 15–44	0.107	0.079	0.113	0.082	0.199
R^2 15–29	0.142	0.056	0.136	0.065	0.225
R^2 30–44	0.058	0.056	0.068	0.057	0.118

Notes: The table shows DID estimates of the pension reform. The dependent variable is a dummy for whether a child was born in the last 12 months. The full set of covariates is as in Table 3, column 6. Rural and urban group definitions are as in Table 2. Mixed urban-rural couples are excluded. Standard errors clustered at the regional level are in parentheses.

Source: PNAD 1981–1990, 1992–1993, 1995–1999

Economic message. This is very hard to square with a purely mechanical wealth story. It fits the paper's argument that sons matter disproportionately as potential heirs and old-age providers.

5.7 Figure 8 and Table 7: long-run fertility and the PAYG fiscal gap

Read:

- Figure 8 shows that the rural-urban completed fertility gap is stable before the reform and then begins to shrink after 1991.
- By 2010 the rural-urban gap in completed fertility has narrowed by about 1.3 births per woman.
- The first year in which the long-run event-study estimate becomes significant is 1994.

- Table 7 translates the fertility decline into a funding-gap calculation for the Brazilian PAYG system. Depending on the assumed administrative cost, the total gap rises from about 0.58–0.59% of GDP in 1995 to about 1.65–1.74% of GDP in 2011.
- The contribution-base effect from lower fertility is much larger than the pure generosity effect. By 2011 it accounts for roughly four-fifths of the total gap.

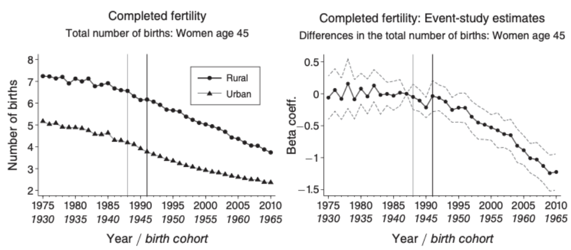

FIGURE 8. COMPLETED FERTILITY: TOTAL BIRTHS AND DIFFERENCES IN TOTAL BIRTHS AT AGE 45

Notes: The left panel shows the average total number of births (alive and dead) of women at age 45. The right panel shows ES estimates of the pension reform for the total number of births (alive and dead) of women at the age of 45 before and after the reform. 90 percent confidence intervals are based on standard errors clustered at the regional (federal state) level. Vertical lines indicate the new constitution approved in 1988 (light gray) and the pension reform implemented in 1991 (black). Rural and urban groups are defined as in Table 2.

Source: PNAD 1984–1985, 1992–1993, 1995–1999, 2001–2009, 2011–2014

TABLE 7—FUNDING GAP OF THE BRAZILIAN PAYG PENSION SYSTEM AS A RESULT OF THE REFORM (IN % OF GDP) FOR DIFFERENT ADMINISTRATIVE COST (*AC*) ASSUMPTIONS

	Generosity effect			Contribution base effect			Total financial gap		
	Administrative cost (*AC*)			Administrative cost (*AC*)			Administrative cost (*AC*)		
	7%	10.75%	14.5%	7%	10.75%	14.5%	7%	10.75%	14.5%
1995	0.07	0.08	0.08	0.51	0.51	0.51	0.58	0.58	0.59
2001	0.12	0.12	0.12	0.74	0.75	0.76	0.86	0.87	0.89
2005	0.18	0.19	0.19	0.86	0.87	0.89	1.03	1.05	1.07
2011	0.34	0.35	0.37	1.31	1.34	1.37	1.65	1.69	1.74

Notes: The table shows calculations of the long-term financial gap of the Brazilian PAYG pension system as a result of the reform (in percent of GDP as of 2012) decomposed into a generosity effect and a contribution base effect. The long-term loss in rural payroll contributors is computed as the difference between the observed number of children and the hypothetical number of children if completed fertility were 1.3 children higher per rural woman. Average pension payments and pension recipients are taken from the PNAD. Pension scheme contributions are monthly average gross income × 10% (employees' average contribution rate) + monthly average gross income × 28% (employers' average contribution rate). Average gross income is average net income + average net income × 18% (the average marginal tax rate).

Source: PNAD 1981–1990, 1992–1993, 1995–1999, 2001–2009, 2011–2014

Economic message. The demographic response is large enough to matter fiscally. Public pensions reduce the supply of future contributors even as they increase current and future benefit claims.

5.8 Robustness facts that matter for interpretation

Two additional findings deserve emphasis even though they are not the core DID tables.

First, the paper finds **no increase in childlessness**. That means the reform mainly compressed higher birth parities rather than making women abandon fertility altogether.

Second, the fertility decline appears to have been implemented largely through **female sterilization**. In the decade after 1986, sterilization rose by about 33% in urban areas and 104% in rural areas, with the age of sterilization falling more in rural areas.

Economic message. The behavioral response was not just latent intention. Women adopted a durable fertility-control technology that made the decline persistent.

6 Short summary

- Brazil's 1991 pension reform sharply increased pension wealth for rural workers, especially women, while leaving urban workers much less affected.
- The paper uses this rural-urban differential shock in a DID design, a 2SLS design, and a long-run event study.
- In the short run, the annual probability of childbirth falls by about 8% overall and by about 24% for women ages 30–44.
- The effect is strongest at higher birth parities and among women who already have sons.
- In the long run, completed fertility at age 45 falls by about 1.3 children for rural women relative to urban women.

- A broad set of confounders—education, health, religion, labor supply, income, wealth, telenovelas, and trade shocks—does not explain the pattern.
- The fertility response is large enough to weaken the future contribution base of the pay-as-you-go pension system.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a large, plausibly exogenous rise in public pension wealth reduces fertility. In Brazil's 1991 reform, the effect is immediate for older women, concentrated on higher-order births, stronger for mothers with sons, and large enough to materially reduce completed fertility in the long run.

7.2 Economic intuition

Children are partly an old-age asset in environments with weak formal insurance. When the state suddenly offers credible and generous pensions, the value of extra children as informal insurance falls. Older women respond most because they are closest to retirement and closest to their desired stopping point. Higher-parity and son-related effects are exactly what one would expect if fertility partly reflected insurance and inheritance motives rather than pure consumption value from children.

7.3 Takeaway

This paper links social insurance to demography using a clean quasi-experiment. The main lesson is not just that pensions reduce fertility; it is that public insurance and private family insurance are substitutes. Once that substitution is strong enough, pension reform changes the long-run population structure and the fiscal arithmetic of the pension system itself.